Ampere's Law

Dr. Sampath Kumar

Associate Professor

Department of ECE

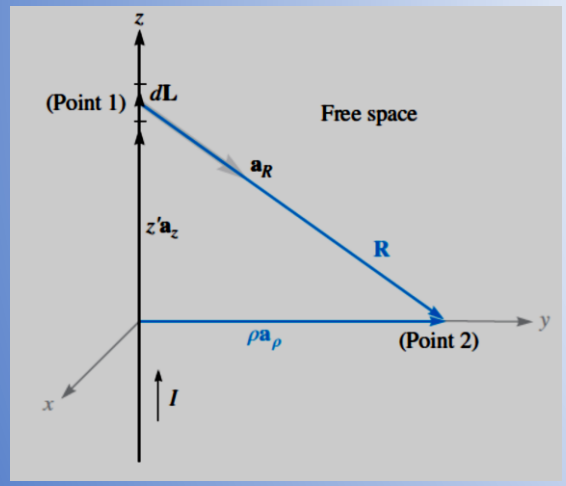
MIT, Manipal

Magnetic flux density due to line charge of finite length at point 2

$$\mathbf{H} = \int_{z_1}^{z_2} \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

$$= \int_{\rho \tan \alpha_1}^{\rho \tan \alpha_2} \frac{I dz \mathbf{a}_z \times (\rho \mathbf{a}_\rho - z\mathbf{a}_z)}{4\pi(\rho^2 + z^2)^{3/2}}$$

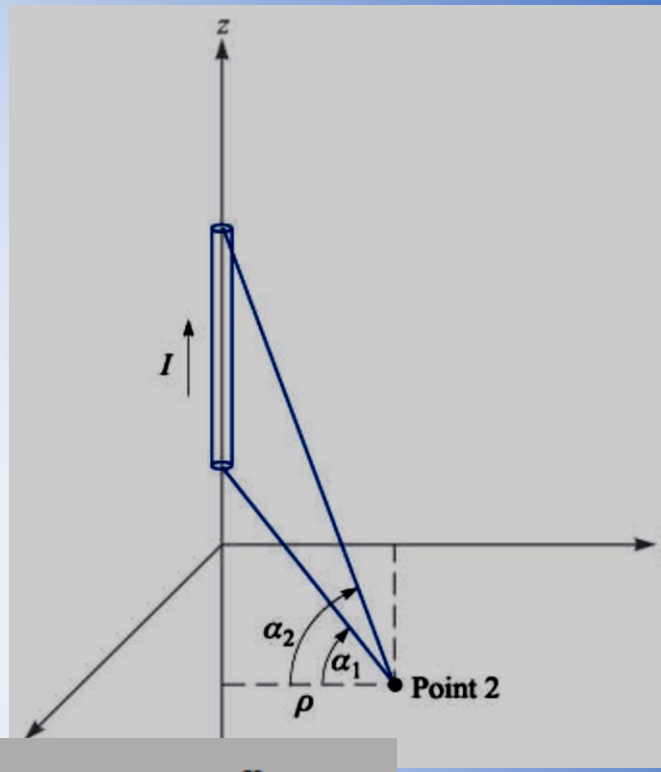
$$= \int_{\rho \tan \alpha_1}^{\rho \tan \alpha_2} \frac{I\rho \mathbf{a}_\phi \, dz}{4\pi(\rho^2 + z^2)^{3/2}}$$



$$= \frac{I z \mathbf{a}_\phi}{4\pi\rho\sqrt{\rho^2 + z^2}} \Bigg|_{\rho \tan \alpha_1}^{\rho \tan \alpha_2}$$

$$= \frac{I}{4\pi\rho}\left[\frac{\tan \alpha_2}{\sqrt{1 + \tan^2 \alpha_2}} - \frac{\tan \alpha_1}{\sqrt{1 + \tan^2 \alpha_1}}\right]\mathbf{a}_\phi$$

$$= \frac{I}{4\pi\rho}(\sin \alpha_2 - \sin \alpha_1)\mathbf{a}_\phi$$



$$I = \int \frac{1}{(x^2 + a^2)^{3/2}} dx = \frac{x}{a^2\sqrt{a^2 + x^2}}$$

Formula

Find H at the center of a square current loop of side L.

Choose a Cartesian coordinate system such that the loop is located as shown in Fig. 10-10. By symmetry, each half-side contributes the same amount to H at the center. For the half-side $0 \le x \le L/2, y = -L/2$, the Biot-Savart law gives for the field at the origin

$$d\mathbf{H} = \frac{(I\,dx\,\mathbf{a}_x) \times [-x\mathbf{a}_x + (L/2)\mathbf{a}_y]}{4\pi [x^2 + (L/2^2)]^{3/2}}$$

$$= \frac{I\,dx\,(L/2)\mathbf{a}_z}{4\pi [x^2 + (L/2^2)]^{3/2}}$$

Therefore, the total field at the origin is

$$\mathbf{H} = 8 \int_0^{L/2} \frac{I\,dx\,(L/2)\mathbf{a}_z}{4\pi [x^2 + (L/2^3)]^{3/2}}$$

$$= \frac{2\sqrt{2}\,I}{\pi L}\mathbf{a}_z$$

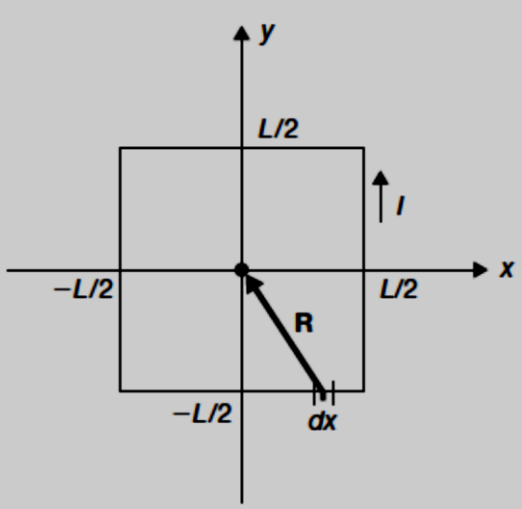


Fig. 10-10

Note: Easier way to solve it –use the formula

$$\frac{I}{4\pi\rho}(\sin\alpha_2 - \sin\alpha_1)\mathbf{a}_\phi$$

And $\alpha = 45^0$

Try this solved example from the text book

EXAMPLE 7.1

As a numerical example illustrating the use of (9), we determine **H** at $P_2(0.4, 0.3, 0)$ in the field of an 8-ampere filamentary current. The current is directed inward from infinity to the origin on the positive x axis, and then outward to infinity along the y axis. This arrangement is shown in Figure 7.6.

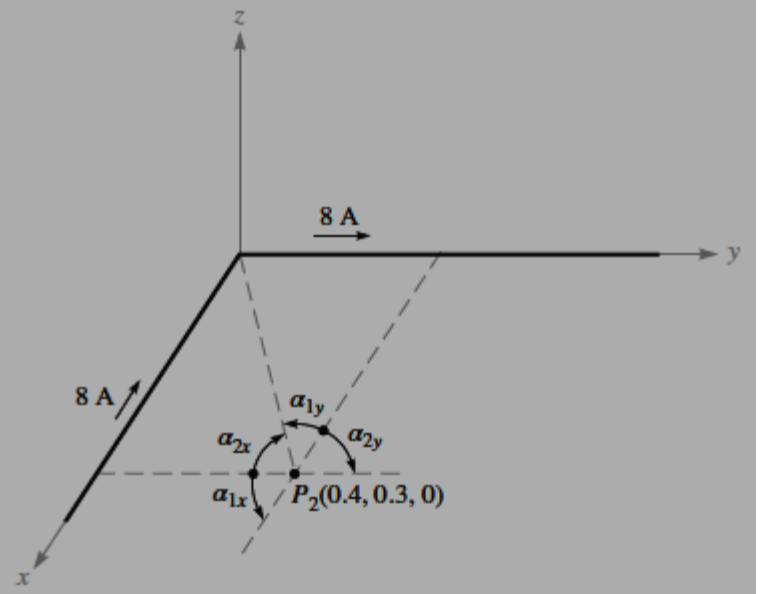


Figure 7.6 The individual fields of two semi-infinite current segments are found by (9) and added to obtain $\mathbf{H}_2$ at P_2.

$$\mathbf{H}_2 = \mathbf{H}_{2(x)} + \mathbf{H}_{2(y)} = -\frac{20}{\pi}\mathbf{a}_z = -6.37\mathbf{a}_z \text{ A/m}$$

Ampere's (Circuital) Law (Similar to Gauss's law)

Ampère's circuital law states that the line integral of **H** about any *closed* path is exactly equal to the direct current enclosed by that path,

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$$

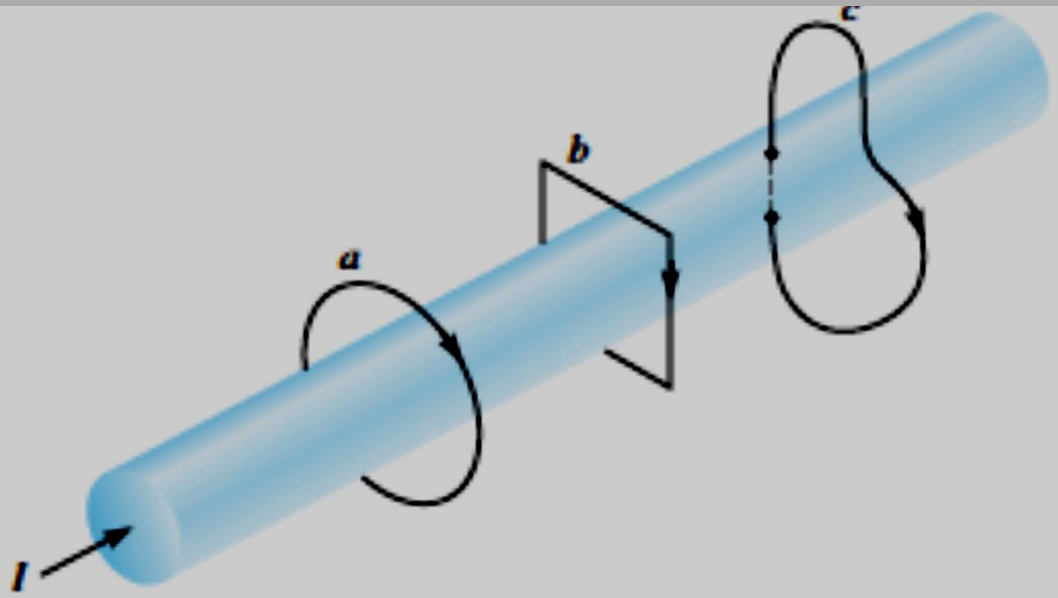


Figure 7.7 A conductor has a total current *I*. The line integral of **H** about the closed paths *a* and *b* is equal to *I*, and the integral around path *c* is less than *I*, since the entire current is not enclosed by the path.

- Find H due to infinitely long current carrying conductor

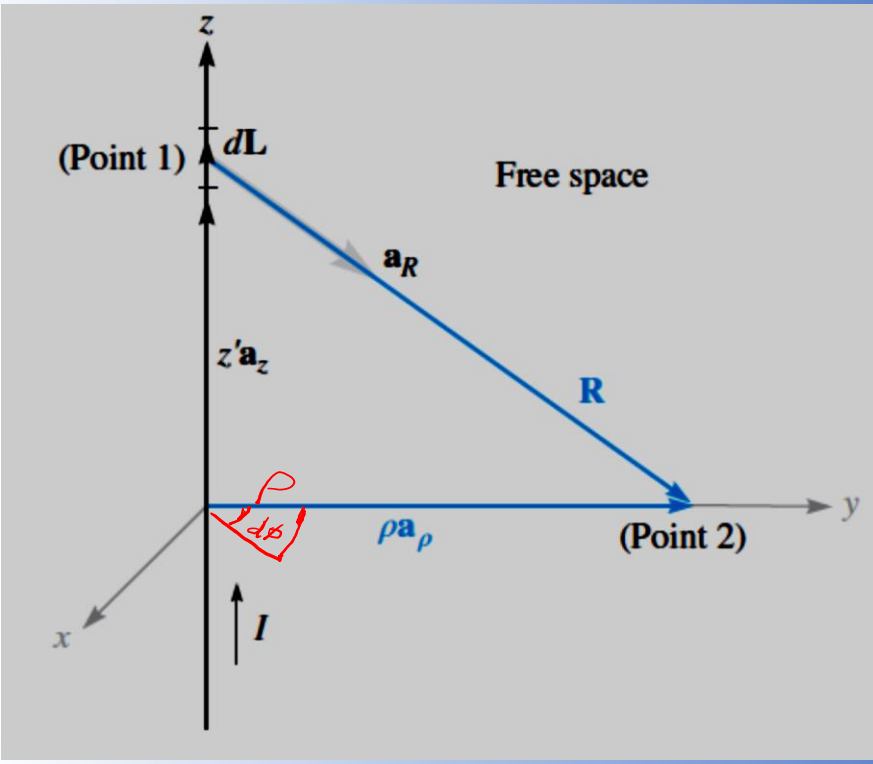
Direction of $d\mathbf{H}$ is perpendicular to the plane containing $d\mathbf{L}$ and $\mathbf{R}$ and therefore is in the direction of $\mathbf{a}_\phi$

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_0^{2\pi} H_\phi \rho d\phi$$

$$= H_\phi \rho \int_0^{2\pi} d\phi$$

$$= H_\phi 2\pi\rho = I$$

$$H_\phi = \frac{I}{2\pi\rho}$$




Thank
you